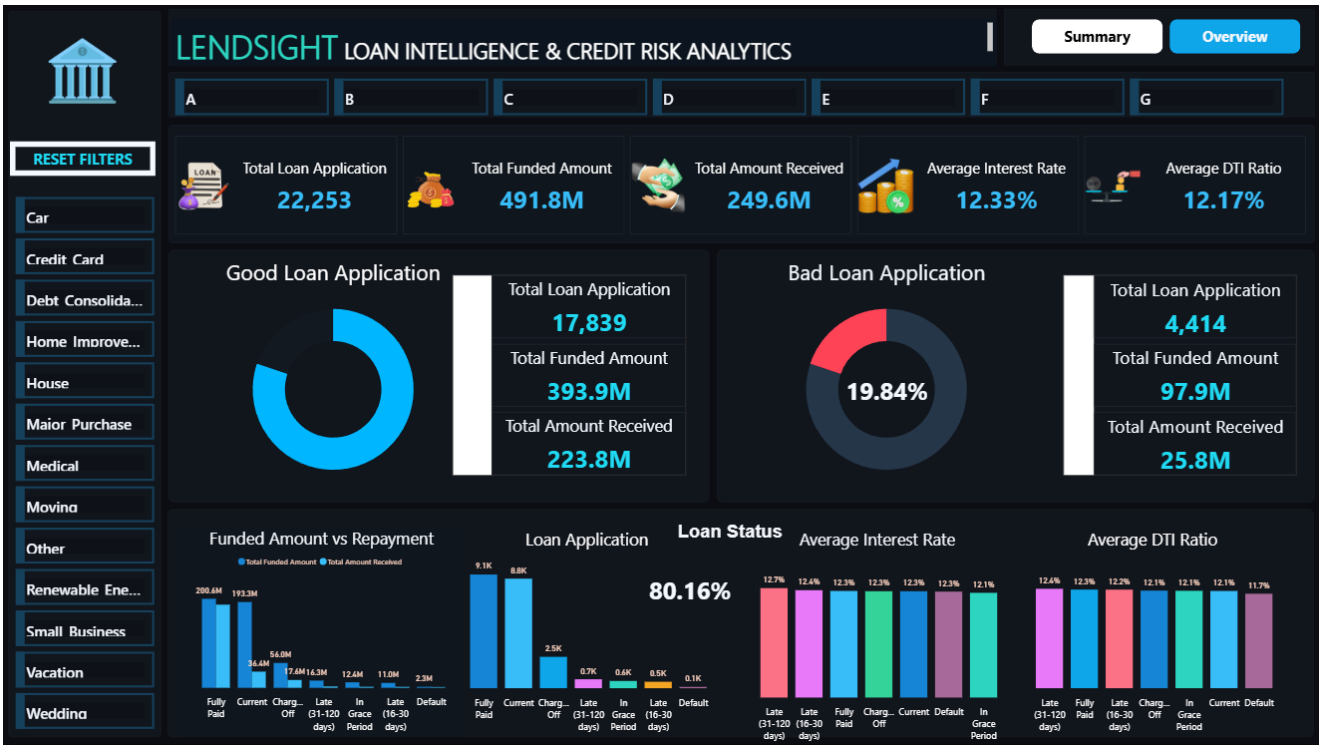
POWER BI PORTFOLIO PROJECT

LendSight

LOAN INTELLIGENCE & CREDIT RISK ANALYTICS

A complete project guide explaining the business problem, dataset, Power BI pages, KPI logic, report interactions, validation process, limitations, and how to present the project to recruiters.

22,253	23	2	2026–2027
synthetic loan records	columns in the workbook	current report pages	synthetic date window



Current Summary page screenshot. Figures vary with active slicer selections; treat displayed values as filtered examples, not fixed project totals.

Important data note

This is a learning and portfolio project built with synthetic data. It is not an actual bank system, and its figures must not be presented as real customer or bank performance.

01 / PROJECT OVERVIEW

Executive summary

What LendSight is, who it is for, and the questions it is designed to answer.

LendSight is an interactive Power BI report for exploring a loan portfolio. It brings application volume, loan funding, borrower repayments, interest rates, debt-to-income (DTI), loan status, purpose, term, employment, home ownership, and geography into a unified visual report.

The current report has two pages: Summary and Overview. It uses a dark black/navy visual style with electric-blue and cyan highlights. Slicers support exploration by loan grade and loan purpose; the Reset Filters button is intended to restore the report to a default state.

Business problem

Loan data is difficult to understand when it is spread across rows and separate calculations. Analysts need a quick way to track portfolio size, funding, amounts received, pricing, borrower leverage, loan statuses, and differences among purposes and borrower segments.

Project objectives

- Summarise loan application volume and principal funded.
- Compare amount funded with amount received as a high-level repayment view.
- Monitor average interest rate and average DTI.
- Compare status groups and distinguish settled loans from loans still active or delinquent.
- Explore monthly activity, term, home ownership, state codes, and loan purposes.
- Provide report navigation and slicers that allow a reviewer to filter the same dataset interactively.

Intended users

User / team	How the report may help
Portfolio analyst	Review volume, funding, receipt totals, status mix, and changes over time.
Credit / risk analyst	Explore grade, interest-rate, DTI, and status patterns for further investigation.
Operations manager	Compare loan-purpose segments and monitor high-level portfolio activity.
Recruiter / interviewer	Assess the creator's data preparation, modelling, DAX, visual design, and analytical communication.

Scope boundary

LendSight is an analytical demonstration, not a credit decision engine. It does not approve or reject real applications and should not be used to make lending decisions.

02 / DATA FOUNDATION

Dataset and preparation

The workbook that powers the current report.

Property	Current project detail
Workbook	LendSight_Synthetic_Loan_Data_22253_Indian_States.xlsx
Worksheet / Power BI table	financial_loan
Rows	22,253 data records (excluding header)
Columns	23
Date columns	ISSUE_DATE, LAST_CREDIT_PULL_DATE, LAST_PAYMENT_DATE, NEXT_PAYMENT_DATE
Date years	Issue dates use 2026 and 2027; the other three date fields also use 2026 and 2027.
State field	ADDRESS_STATE contains 36 two-letter Indian state / UT-style codes.
Data status	Synthetic records generated for learning and demonstration, not real customer records.

Data preparation workflow

1. Open the workbook and check that the financial_loan worksheet is selected.
2. Connect from Power BI Desktop using Home → Excel workbook, or update the existing source through Power Query.
3. Confirm all 23 column names are preserved and expected fields have correct types. For MTD/MoM analysis, create a continuous Calendar table, mark it as the date table, and relate Calendar[Date] to ISSUE_DATE.
4. Set the four date columns to Date, numeric amounts to Decimal Number / Fixed Decimal as appropriate, and categorical fields to Text.
5. Check that INT_RATE and DTI are stored as ratios in this workbook (for example, 0.1320 represents 13.20%). Format these as percentages in Power BI.
6. Refresh the model, inspect visuals, verify slicer interactions, and save the PBIX.

Important data interpretation notes

- ADDRESS_STATE values were randomly assigned from an Indian state / Union Territory code list for visual practice. They are not validated borrower locations and must not be used to infer genuine Indian regional loan patterns.
- The synthetic date range was shifted to 2026–2027. Future dates in this practice dataset are intentional; they do not describe real lending activity.
- Currency units are not explicitly documented in the workbook. Decide on and display a consistent currency/unit label before publishing any report.
- TOTAL_PAYMENT is simulated. It is not an audited repayment ledger and should not be used to claim actual collection performance.

03 / DATA DICTIONARY

Field reference: identifiers, categories and dates

Use this to understand what each column represents and where it may be used in the report.

Field	Type	Meaning / use
ISSUE_DATE	Date	Date the loan application/loan record is issued; main time-series field.
ID	Identifier	Synthetic row / loan identifier; useful for uniqueness checks.
PURPOSE	Category	Purpose such as debt_consolidation, credit_card, car, house, etc.
VERIFICATION_STATUS	Category	Income / information verification category: Verified, Source Verified, Not Verified.
GRADE	Category	Loan grade A–G; used by the A–G slicer.
HOME_OWNERSHIP	Category	Borrower housing category, e.g. MORTGAGE, RENT, OWN.
ADDRESS_STATE	Category	Two-letter synthetic Indian state / UT code assigned for map/chart practice.
LOAN_STATUS	Category	Repayment / account state, e.g. Fully Paid, Current, Charged Off, late states, grace period, Default.
EMP_LENGTH	Category	Reported length of employment; segment for employment analysis.
EMP_TITLE	Category / text	Employment title; potentially high-cardinality detail field.
LAST_CREDIT_PULL_DATE	Date	Synthetic date for most recent credit pull.
LAST_PAYMENT_DATE	Date	Synthetic date for most recent payment.
NEXT_PAYMENT_DATE	Date	Synthetic next-payment date.

Data modelling hint

Keep ID and MEMBER_ID as identifiers, not numeric measures. A sum of identifiers has no business meaning. Use COUNTROWS or DISTINCTCOUNT when counting records or unique IDs.

03 / DATA DICTIONARY

Field reference: loan and borrower measures

These fields support funding, repayment, pricing, and borrower profile metrics.

Field	Type	Meaning / use
MEMBER_ID	Identifier	Synthetic borrower/member identifier; can support member-level analysis if uniqueness and relationships are validated.
SUB_GRADE	Category	More detailed loan grade, such as C5 or B4.
TERM	Category	Loan duration: 36 months or 60 months.
ANNUAL_INCOME	Numeric amount	Synthetic annual borrower income. Currency unit should be explicitly stated before publishing.
DTI	Ratio	Debt-to-income ratio. Example 0.1656 is 16.56%; format as Percentage.
INSTALLMENT	Numeric amount	Scheduled installment amount; monetary unit not explicitly confirmed.
INT_RATE	Ratio	Interest rate stored as a decimal ratio. Example 0.1320 is 13.20%; format as Percentage.
LOAN_AMOUNT	Numeric amount	Loan principal / funded amount represented by a record.
TOTAL_ACC	Count	Synthetic total-account count for a borrower record.
TOTAL_PAYMENT	Numeric amount	Synthetic amount received / paid field; not audited actual repayments.

Observed categories in the supplied workbook

Field	Observed values / profile
LOAN_STATUS	7 values: Fully Paid, Current, Charged Off, Late (31-120 days), In Grace Period, Late (16-30 days), Default.
PURPOSE	13 values: debt_consolidation, credit_card, home_improvement, major_purchase, car, medical, small_business, house, moving, vacation, other, wedding, renewable_energy.
TERM	36 months and 60 months.
GRADE	A, B, C, D, E, F, G.
HOME_OWNERSHIP	MORTGAGE, RENT, OWN, OTHER, ANY, NONE.
VERIFICATION_STATUS	Verified, Source Verified, Not Verified.
ADDRESS_STATE	36 state / UT-style two-letter codes, synthetic/random assignment.

04 / REPORT WALKTHROUGH

Page 1 — Summary

A compact health check of the selected loan portfolio.

Dashboard image reference

A full-size Summary screenshot appears in the visual appendix at the end of this PDF. Values shown in the report change with active grade and purpose filters.

Top KPI cards

KPI	Business meaning	Recommended calculation
Total Loan Applications	Number of records/applications in the current filter context.	COUNTROWS(financial_loan), or DISTINCTCOUNT(ID) if IDs are verified unique.
Total Funded Amount	Sum of principal / loan amount.	SUM(LOAN_AMOUNT).
Total Amount Received	Sum of the supplied payment field.	SUM(TOTAL_PAYMENT); interpret as simulated.
Average Interest Rate	Average quoted rate in the filtered records.	AVERAGE(INT_RATE), formatted as percentage.
Average DTI Ratio	Average debt-to-income ratio.	AVERAGE(DTI), formatted as percentage.

Good loan vs Bad loan panels

The report uses doughnut charts for a quick comparison and supporting values for applications, funded amount, and amount received. The current dashboard grouping is consistent with the supplied dataset counts: Fully Paid (9,085) + Current (8,754) = 17,839 Good applications; all other statuses total 4,414 Bad applications. This is a project-specific grouping, not a universal banking standard. In Grace Period and late statuses require a clearly documented business rule if this were a production credit-risk report.

Loan Status strip

The bottom row compares funded vs received amounts and displays counts, average interest rate and average DTI by loan status. It is designed to help a reviewer compare statuses, rather than relying on one total alone.

04 / REPORT WALKTHROUGH

Page 2 — Overview

Trends and segmentation: when loans were issued and how the records are distributed.

Dashboard image reference

A full-size Overview screenshot appears in the visual appendix at the end of this PDF. Note that the map visual is disabled in the supplied screenshot; enable it or replace it with a state-code bar chart before final publication.

Visual	What it answers	How to interpret it
Monthly Loan Applications	How application volume changes by issue month.	Use a date axis that separates years; January 2026 and January 2027 must not be combined unintentionally.
Loan Applications by State	Where records are distributed.	Current map visual is disabled in the screenshot. Because state codes are randomly assigned, use only as a demonstration or replace with a bar chart.
Loan Applications by Term	How 36-month and 60-month terms are represented.	Compare counts or share by term; confirm the chart legend matches the field values.
Home Ownership	How applications distribute across housing categories.	The largest bars show the most common categories in the current filter context.
Loan Purpose	Which stated reasons are most common.	A treemap shows relative counts/amounts; label the value field used in the visual.

Map alternative

If Power BI maps remain disabled, replace the filled map with a horizontal bar chart: Axis = ADDRESS_STATE; Values = Total Loan Applications or Total Funded Amount. Clearly label it as a synthetic state-code demonstration.

05 / ANALYTICS LOGIC

KPI definitions and sample DAX

These examples are recommended definitions. Compare them with your existing measures before replacing anything.

The formulas below use the current Power BI table name financial_loan and the workbook's uppercase column names. If your model table has a different name, update the formulas accordingly.

Core portfolio measures

```
Total Loan Applications =
COUNTROWS(financial_loan)

Total Funded Amount =
SUM(financial_loan[LOAN_AMOUNT])

Total Amount Received =
SUM(financial_loan[TOTAL_PAYMENT])

Average Interest Rate =
AVERAGE(financial_loan[INT_RATE])

Average DTI Ratio =
AVERAGE(financial_loan[DTI])
```

Status measures with a transparent rule

```
Good Loan Applications =
CALCULATE(
    [Total Loan Applications],
    financial_loan[LOAN_STATUS] IN {"Fully Paid", "Current"}
)

Bad Loan Applications =
CALCULATE(
    [Total Loan Applications],
    NOT (financial_loan[LOAN_STATUS] IN {"Fully Paid", "Current"})
)

Good Loan % =
DIVIDE([Good Loan Applications], [Total Loan Applications])

Bad Loan % =
DIVIDE([Bad Loan Applications], [Total Loan Applications])
```

These examples match the current screenshot grouping: Fully Paid and Current are grouped as Good; Charged Off, late, In Grace Period, and Default are grouped as Bad. Percentages use all applications as the denominator. This is a project-defined rule for synthetic data, not a universal banking risk policy. Validate the business definition before using such labels in a real institution.

Time intelligence: prerequisites

```
Calendar =
CALENDAR(
    MIN(financial_loan[ISSUE_DATE]),
    MAX(financial_loan[ISSUE_DATE])
)

Loan Applications MTD =
TOTALMTD([Total Loan Applications], 'Calendar'[Date])
```

06 / USER EXPERIENCE

Model, filters and navigation

How the report is intended to behave when a viewer explores the data.

Data flow

Excel workbook	Power Query	Data model	DAX measures	Power BI pages
23 columns	Types / checks	Date relationship	KPIs / ratios	Charts / slicers

The current report is connected to the financial_loan table. A mature model should keep descriptive columns as categories, numeric measures as numeric types, and the date dimension separate for time intelligence.

Filters and interactions

- A–G slicer: filters records by GRADE.
- Purpose slicer: filters by PURPOSE such as car, credit_card, debt_consolidation, etc.
- Summary / Overview buttons: navigate between report pages; actions may require Ctrl+click inside Power BI Desktop editing mode.
- Reset Filters button: uses a Reset Filters bookmark saved with the desired unfiltered state. In Desktop editing view, hold Ctrl while clicking the button to activate its action.
- Bookmark reset state: clear slicers first, then update the bookmark. Make sure Data is enabled and the bookmark targets all relevant visuals.

Design system

Use	Hex	Design intent
Page / canvas background	#05070B	Deep black, reduces glare and supports contrast.
Visual cards	#101720	Separate dashboard panels from the canvas.
Primary blue	#009DFF	Main bar/selected-state emphasis.
Bright cyan	#20D9F2	KPI values and active highlights.
Good loans	#00B7FF	Positive/performing segment accent.
Bad loans / risk	#FF4357	Risk signal; keep its use consistent.
Border	#16415C	Subtle outline around panels and cards.
Main text	#F4F7FB	Readable labels and titles.

07 / QUALITY AND CAVEATS

Validation checklist and limitations

Complete these checks before publishing to GitHub or describing the project in an interview.

Data validation checklist

- ☐ Confirm row count = 22,253 after refresh and confirm the table has 23 columns.
- ☐ Check ID uniqueness before using DISTINCTCOUNT(ID) as the application count.
- ☐ Confirm ISSUE_DATE and other date columns are Date types and fall in the intended synthetic date window.
- ☐ Verify NEXT_PAYMENT_DATE is not before ISSUE_DATE for all records; check last-payment dates and credit-pull dates for plausible chronology.
- ☐ Format INT_RATE and DTI correctly as percentages (the data stores examples such as 0.1320 and 0.1656).
- ☐ Confirm amounts use the intended unit and display label; do not assume INR if not defined.
- ☐ Verify the Good / Bad loan classification and percentage denominator. Do not classify Current and delinquent statuses inconsistently.
- ☐ Clear all slicers and verify global KPIs; then select a category and check that all affected visuals respond as intended.
- ☐ Test Reset Filters with different combinations of GRADE and PURPOSE, using Ctrl+click in Desktop editing mode.
- ☐ Fix or replace the disabled Azure Maps visual on Overview.
- ☐ Standardise the dashboard title to LendSight: the supplied Summary screenshot has a visible spelling variant compared with the Overview screenshot.
- ☐ Fix the white vertical strips shown beside the Good/Bad loan supporting KPI stacks in the Summary screenshot before using it as a final portfolio image.
- ☐ Inspect for blank visuals, truncated titles, illegible labels, and overlapping chart elements at normal report zoom.

Limitations to disclose

Limitation	Why it matters
Synthetic dataset	No real customer-level conclusions; sample values are for demonstrating analytics only.
Synthetic geography	State labels were randomly assigned; geography is not evidence of actual borrower distribution.
Synthetic future dates	2026–2027 dates were generated for practice; monthly charts are not historical market evidence.
Simulated payment total	TOTAL_PAYMENT is not an audited loan repayment history.
Currency ambiguity	Workbook does not clearly specify currency in the field names; define units before publication.
Loan status rules	Good/bad classification must match a stated business definition and a tested DAX denominator.
Map currently disabled	Current screenshot reports mapping disabled by Power BI security/settings; replace it or configure it before demo.

08 / ANALYSIS AND ENHANCEMENTS

How to turn charts into insights

Use these questions to guide a review; do not claim conclusions until the refreshed report supports them.

Questions LendSight can help explore

- Which loan purposes account for the highest application volume and the highest funded amount? These rankings may differ.
- How does the loan-status mix vary across grades A–G?
- Do average interest rate and DTI differ across loan grades or statuses?
- What share of records uses 36-month versus 60-month terms?
- How do application counts and funded amounts vary by home-ownership category and employment length?
- Does monthly application volume change between 2026 and 2027 after the year-month axis is corrected?
- How does amount received compare with amount funded by status? Treat this as a descriptive ratio only because payments are simulated.

Useful enhancements

Priority	Enhancement	Benefit
High	Add a full Date table and Year-Month axis.	Avoids combining same-named months across 2026 and 2027.
High	Standardise Good / Bad / Current definitions.	Prevents misleading risk and repayment percentages.
High	Replace disabled map with a state-code bar chart or configure mapping intentionally.	Ensures Overview has no broken/blank visual.
Medium	Add a Details page with searchable loan records and controlled drill-through.	Gives a detailed view without cluttering Summary.
Medium	Add MTD, previous-month, and MoM-change KPIs with a verified date relationship.	Makes trend reporting more useful.
Medium	Add tooltips and a consistent currency label.	Improves interpretation and presentation quality.
Later	Connect an SQL Server database and reproduce selected metrics in SQL.	Demonstrates cross-tool validation if schema is mapped carefully.

Recommended publication standard

Label this as a synthetic analytics case study. Distinguish demonstrated functionality from future enhancements, and avoid presenting generated dates or state assignments as real lending insights.

09 / PORTFOLIO AND INTERVIEW

How to explain LendSight

A concise, honest way to describe the project to a recruiter or interviewer.

60-second project explanation

Sample pitch

"LendSight is a Power BI portfolio project that explores a synthetic loan portfolio. I prepared a 22,253-row, 23-column dataset and designed two report pages: Summary and Overview. The Summary page tracks applications, funded amounts, simulated amounts received, average interest rate, DTI, and loan-status metrics. The Overview page explores monthly trends, loan terms, home ownership, purpose and state-code distributions. I used slicers, a reset bookmark, and a black-and-electric-blue design to make the report interactive. Because the data is synthetic, I treat the dashboard as a demonstration of analytics skills rather than evidence about a real bank."

Suggested résumé project entry

LendSight — Loan Intelligence & Credit Risk Analytics | Power BI, Excel, DAX

- Built a two-page interactive Power BI report over a 22,253-row synthetic loan dataset with 23 fields.
- Created KPI measures for loan applications, funded principal, simulated amount received, average interest rate and DTI.
- Designed loan-status, term, purpose, home-ownership and monthly-trend visuals with grade/purpose slicers and a reset bookmark.
- Applied a consistent dark navy, electric-blue and cyan visual system and documented data-quality checks and model limitations.

Interview questions to practise

Question	Strong answer should cover
Why did you build this?	Business need, loan portfolio visibility, intended analyst/operations audience.
How did you calculate total funded amount?	SUM of LOAN_AMOUNT; explain grain and currency/unit caveat.
What is DTI?	Debt-to-income ratio; explains debt burden relative to income, but avoid using synthetic records for real credit decisions.
How does the reset button work?	Bookmark saves a known default state; button action points to bookmark; Ctrl+click triggers it in Desktop edit mode.
How did you validate the report?	Row/column counts, data types, date ranges, calculations, slicer interactions, reset action, disabled-map replacement.
What are the limitations?	Synthetic values, random state mapping, 2026–2027 dates, simulated total payment, need for verified status definitions.

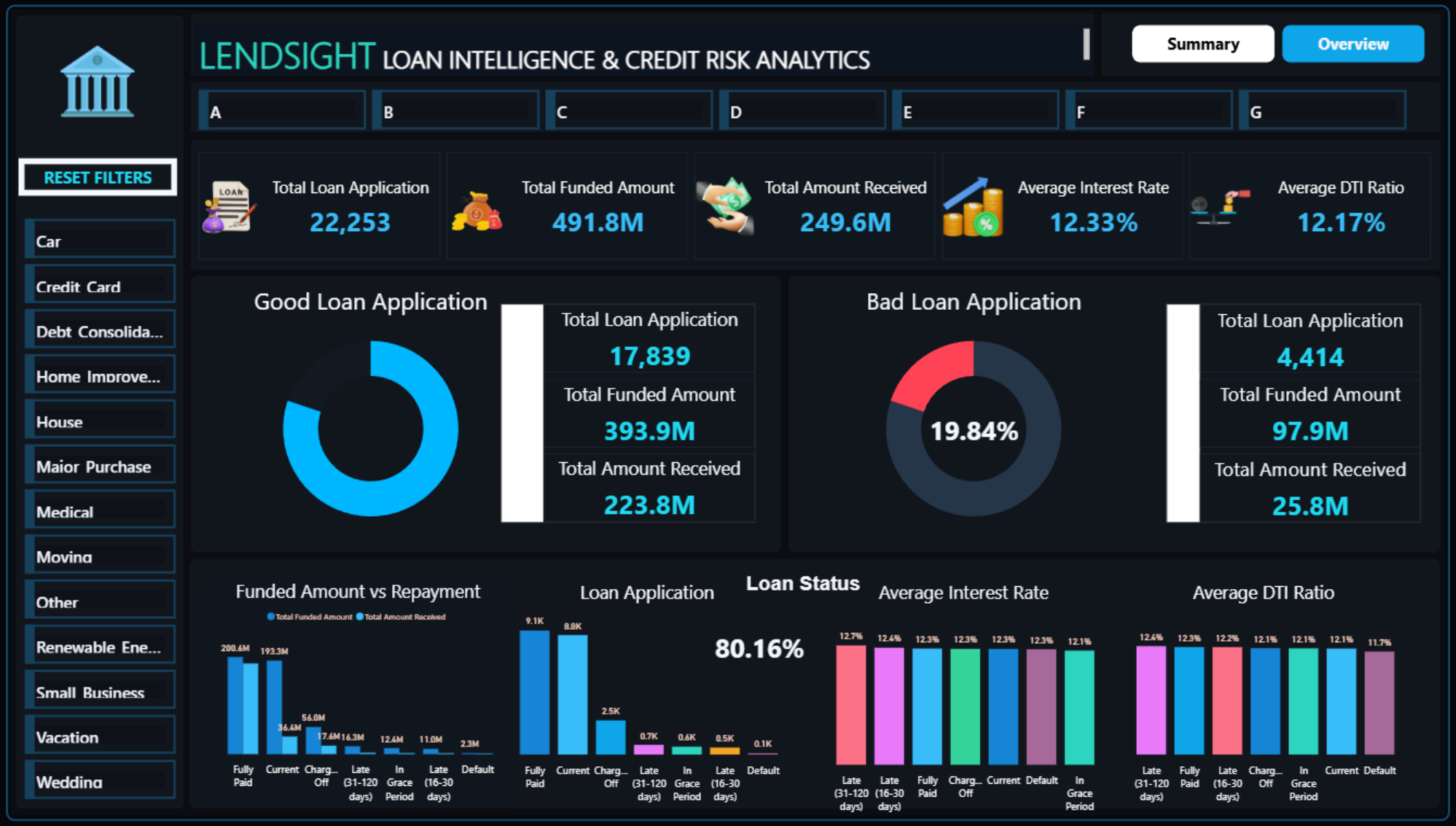
Final checklist before sharing

- ☐ Rename report title consistently to LendSight (not LENDLIGHT).
- ☐ Save the PBIX and keep a backup of the workbook used as source.
- ☐ Capture clean screenshots of both pages with slicers cleared.
- ☐ Publish a README including objective, dataset note, KPIs, visuals, tools, limitations, and screenshots.
- ☐ Never upload real personal or confidential banking data.

Prepared as a project explanation guide based on the supplied LendSight workbook, project brief, SQL starter material, and dashboard screenshots.

Appendix A - Summary Dashboard

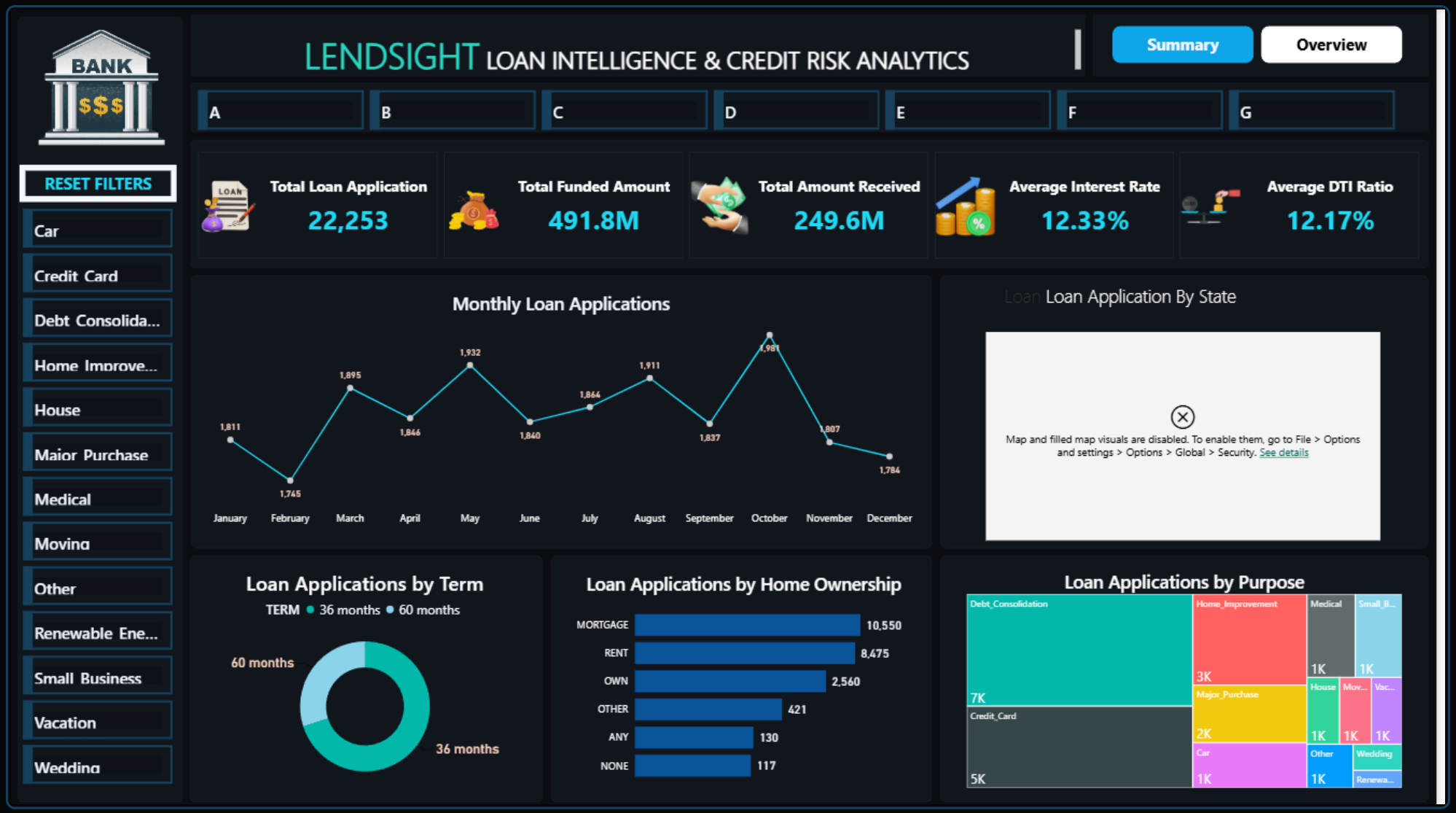
Full-size supplied screenshot | Summary page | Image is shown without content retouching.



Review note: the supplied Summary screenshot shows white vertical strips beside the supporting Good/Bad loan KPI cards. Remove these in Power BI before public portfolio sharing.

Appendix B - Overview Dashboard

Full-size supplied screenshot | Overview page | Image is shown without content retouching.



Review note: the state map is disabled in the supplied screenshot. Enable map visuals or replace the map with a state-code bar chart; state assignments in this dataset are synthetic.